# A SHEAF GLUED FROM OPEN ALGEBRAIC SUBFUNCTORS IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The statement carries a nested enumerate with an internal reference to the definition of a relative representable property; both references stay inside the counted statement span.

*Editorial note.* The nine references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

*Allowed standard prerequisites (editorial).*

- Representability of the diagonal is tested by fibre products of maps from schemes. Lemma lemma-representable-diagonal (921-934), used at proof lines 1336-1340. Reduces representing the diagonal to representing G = T x_F T'.
- Gluing of representable functors along a Zariski covering. Schemes, Lemma schemes-lemma-glue-functors, used at proof lines 1344-1346. Produces representability of G from the G_i.
- Fibre products of sheaves with coproducts of subfunctors. Lemma lemma-representable-sheaf-coproduct-sheaves (1228-1236), used at proof lines 1350-1352. Writes U as the coproduct of the U_i.
- Composition of representable transformations with a property. Lemma lemma-composition-representable-transformations-property, used at proof lines 1353-1356. Gives surjectivity of U -> F.
- Open immersions and compositions are etale. Morphisms, Lemmas morphisms-lemma-open-immersion-etale and morphisms-lemma-composition-etale, used at proof lines 1360-1364. Gives etaleness of U_i x_F T -> T.

*Source reference.* Original definition 025V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2828–2828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02WK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3807–3807. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3813–3813. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 025W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2829–2829. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02WO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3811–3811. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3234–3234. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02GP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3236–3236. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original remark 02YN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3882–3882. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 01JJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2016–2016. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $S \in \mathrm{Ob}(Sch_{fppf})$. Let $F$ be a presheaf of sets on $(Sch/S)_{fppf}$. Assume*

1. *$F$ is a sheaf,*
2. *there exists an index set $I$ and subfunctors $F_i \subset F$ such that*
   (a) *each $F_i$ is an algebraic space,*
   (b) *each $F_i \to F$ is representable,*
   (c) *each $F_i \to F$ is an open immersion (see Definition 025V),*
   (d) *the map $\coprod F_i \to F$ is surjective as a map of sheaves, and*
   (e) *$\coprod F_i$ is an algebraic space (set theoretic condition, see Lemma 02WQ).*

*Then $F$ is an algebraic space.*

*Proof.* Let $T$ be an object of $(Sch/S)_{fppf}$. Let $T \to F$ be a morphism. By assumption (2)(b) and (2)(c) the fibre product $F_i \times_F T$ is representable by an open subscheme $V_i \subset T$. It follows that $(\coprod F_i) \times_F T$ is represented by the scheme $\coprod V_i$ over $T$. By assumption (2)(d) there exists an fppf covering $\{T_j \to T\}_{j \in J}$ such that $T_j \to T \to F$ factors through $F_i$, $i = i(j)$. Hence $T_j \to T$ factors through the open subscheme $V_{i(j)} \subset T$. Since $\{T_j \to T\}$ is jointly surjective, it follows that $T = \bigcup V_i$ is an open covering. In particular, the transformation of functors $\coprod F_i \to F$ is representable and surjective in the sense of Definition 025V (see Remark 02YN for a discussion).

Next, let $T' \to F$ be a second morphism from an object in $(Sch/S)_{fppf}$. Write as above $T' = \bigcup V_i'$ with $V_i' = T' \times_F F_i$. To show that the diagonal $F \to F \times F$ is representable we have to show that $G = T \times_F T'$ is representable, see Lemma

025W. Consider the subfunctors $G_i = G \times_F F_i$. Note that $G_i = V_i \times_{F_i} V_i'$, and hence is representable as $F_i$ is an algebraic space. By the above the $G_i$ form a Zariski covering of $G$. Hence by Schemes, Lemma 01JJ we see $G$ is representable.

Choose a scheme $U \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective étale morphism $U \to \coprod F_i$ (this exists by hypothesis). We may write $U = \coprod U_i$ with $U_i$ the inverse image of $F_i$, see Lemma 02WO. We claim that $U \to F$ is surjective and étale. Surjectivity follows as $\coprod F_i \to F$ is surjective (see first paragraph of the proof) by applying Lemma 02WK. Consider the fibre product $U \times_F T$ where $T \to F$ is as above. We have to show that $U \times_F T \to T$ is étale. Since $U \times_F T = \coprod U_i \times_F T$ it suffices to show each $U_i \times_F T \to T$ is étale. Since $U_i \times_F T = U_i \times_{F_i} V_i$ this follows from the fact that $U_i \to F_i$ is étale and $V_i \to T$ is an open immersion (and Morphisms, Lemmas 02GP and 02GN). □